ACR Accreditation Scan Measurements

Complete the form entering values for all measurements. Type data directly into the form or print the form and use it as paper form. The term "slice" refers to space locations, not necessarily the image number.

Site Name:		Date (Sample Format: 31-May-2012):	
System Configuration:			
Software Version:			
Field Strength:			
Table Type (Flat or Curved):			
Tester Name:			
ACR Phantom Serial Number:			
Head-Type Coil:			
Head-Type Coil Model #:		Head-Type Coil Serial #:	
Patient Weight:			

Image Information		
ACR Localizer	Exam	
	Series	
	Images	
	Center Frequency	
	AutoPreScan TG	
	Actual TG Used	
	R1	
	R2	
ACR T1	Exam	
	Series	
	Images	
	Center Frequency	
	AutoPreScan TG	
	Actual TG Used	
	R1	
	R2	
ACR T2	Exam	
	Series	
	Images	
	Center Frequency	
	AutoPreScan TG	
	Actual TG Used	
	R1	
	R2	

Geometric Accuracy		
ACR Sagittal Localizer	Reference Window Level	
	S/I Length	mm
	S/I Length large phantom: 148 mm ± 3 mm	☐Yes ☐No
	S/I Length medium phantom: 134 mm ± 2.0 mm	☐Yes ☐No
		☐ **Pass** ☐ **Fail**
ACR T1 (slice 1)	Reference Window Level	
	Right/Left (R/L) diameter	mm
	Anterior/Posterior (A/P) diameter	mm
ACR T1 (slice 5)	Reference Window Level	
	Right/Left (R/L) diameter	mm
	Anterior/Posterior (A/P) diameter	mm
	+45 degree diagonal diameter (pos. slope)	mm
	-45 degree diagonal diameter (neg. slope)	mm
Each of the 6 diameters is equal to:	Large phantom: 190 mm ± 3 mm	☐Yes ☐No
	Medium phantom: 165 mm ± 2.0 mm	☐Yes ☐No
		☐ **Pass** ☐ **Fail**
High Contrast Spatial Resolution		
ACR T1 (slice 1) 1st Array Pair	Left-Right resolved @1.1 mm	☐Yes ☐No
	Top-Bottom resolved @1.1 mm	☐Yes ☐No
ACR T1 (slice 1) 2nd Array Pair	Left-Right resolved @1.0 mm	☐Yes ☐No
	Top-Bottom resolved @1.0 mm	☐Yes ☐No
ACR T1 (slice 1) 3rd Array Pair	Left-Right resolved @0.9 mm	☐Yes ☐No
	Top-Bottom resolved @0.9 mm	☐Yes ☐No
ACR T1 (slice 1) 4th Array Pair (if applicable)	Left-Right resolved @0.8 mm	☐Yes ☐No
	Top-Bottom resolved @0.8 mm	☐Yes ☐No
ACR T2 (slice 1, 2nd echo) 1st Array Pair	Left-Right resolved @1.1 mm	☐Yes ☐No
	Top-Bottom resolved @1.1 mm	☐Yes ☐No
ACR T2 (slice 1, 2nd echo) 2nd Array Pair	Left-Right resolved @1.0 mm	☐Yes ☐No
	Top-Bottom resolved @1.0 mm	☐Yes ☐No
ACR T2 (slice 1, 2nd echo) 3rd Array Pair	Left-Right resolved @0.9 mm	☐Yes ☐No
	Top-Bottom resolved @0.9 mm	☐Yes ☐No

ACR T2 (slice 1) 4th Array Pair (if applicable)	Left-Right resolved @0.8 mm	☐Yes ☐No
	Top-Bottom resolved @0.8 mm	☐Yes ☐No
The ACR T1 and ACR T2 images analyzed are resolved in both the Left-Right and Top-Bottom directions to at least 1.0 mm (for example, resolved at both 1.1 mm and 1.0 mm).		☐Yes ☐No
		☐ **Pass** ☐ **Fail**
Slice Thickness Accuracy		
ACR T1 (slice 1)	Top Ramp ROI Mean	
	Bottom Ramp ROI Mean	
	Top Ramp Length	mm
	Bottom Ramp Length	mm
	Slice Thickness	mm
ACR T2 (slice 1, 2nd echo)	Top Ramp ROI Mean	
	Bottom Ramp ROI Mean	
	Top Ramp Length	mm
	Bottom Ramp Length	mm
	Slice Thickness	mm
Each T1 & T2 Slice 1 Thicknesses = 5.0 ± 1.0 mm for both large and medium phantoms		☐Yes ☐No
		☐ **Pass** ☐ **Fail**
Slice Position Accuracy		
ACR T1 (slice 1)	Reference Window Level	
	Bar Length Difference	mm
ACR T1 (slice 11)	Reference Window Level	
	Bar Length Difference	mm
ACR T2 (slice 1, 2nd echo)	Reference Window Level	
	Bar Length Difference	mm
ACR T2 (slice 11, 2nd echo)	Reference Window Level	
	Bar Length Difference	mm
Absolute Value of Bar Length Differences (all 4) ≤ 5.0 mm		☐Yes ☐No
		☐ **Pass** ☐ **Fail**
Image Intensity Uniformity		
ACR T1 (slice 7)	ROI Low Signal Mean	
	ROI High Signal Mean	
	T1 PIU (%)	%
ACR T2 (slice 7, 2nd echo)	ROI Low Signal Mean	
	ROI High Signal Mean	
	T2 PIU (%)	%
	Field Strength (1.5T/3.0T)	T
If 1.5T system:	Large phantom: Each T1 & T2 PIU % ≥ 85.0%	☐Yes ☐No ☐N/A
	Medium phantom: Each T1 & T2 PIU % ≥ 90%	☐Yes ☐No ☐N/A

If 3.0T system:	Large phantom: Each T1 & T2 PIU % ≥ 80.0%	☐Yes ☐No ☐N/A
	Medium phantom: Each T1 & T2 PIU % ≥ 85%	☐Yes ☐No ☐N/A
		☐ **Pass** ☐ **Fail**
Percent-Signal Ghosting		
ACR T1 (slice 7)	ROI Large Signal Mean	
	ROI Top Signal Mean	
	ROI Bottom Signal Mean	
	ROI Left Signal Mean	
	ROI Right Signal Mean	
	Ghosting Ratio	
The ACR T1 Ghosting Ratio is ≤ 0.03 for both large and medium phantoms		☐Yes ☐No
		☐ **Pass** ☐ **Fail**
Low-Contrast Object Detectability		
ACR T1 (slice 11)	Number of Complete Spokes	
ACR T1 (slice 10)	Number of Complete Spokes	
ACR T1 (slice 9)	Number of Complete Spokes	
ACR T1 (slice 8)	Number of Complete Spokes	
ACR T1 Total Spokes (sum of previous 4 entries)		0
ACR T2 (slice 11, 2nd echo)	Number of Complete Spokes	
ACR T2 (slice 10, 2nd echo)	Number of Complete Spokes	
ACR T2 (slice 9, 2nd echo)	Number of Complete Spokes	
ACR T2 (slice 8, 2nd echo)	Number of Complete Spokes	
ACR T2 Total Spokes (sum of previous 4 entries)		0
	Field Strength (1.5T/3.0T)	T
If 1.5T system:	Large and medium phantoms: 30 Spokes for T1, 25 Spokes for T2	☐Yes ☐No ☐N/A
If 3.0T system:	Large and medium phantoms: 37 Spokes for both T1 and T2 across slices 8-11	☐Yes ☐No ☐N/A
		☐ **Pass** ☐ **Fail**

GE HealthCare Representative ______________________ Date ______________